McMasterNLP at KnowledgeGraphEval-2026: Heterogeneous Expert Systems for Arabic Knowledge Graph Construction

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Abstract

Automated knowledge graph (KG) construction involves two core components: nested named entity recognition (NER) and relation extraction (RE). In this work, we address cross-domain nested NER alongside RE. We integrate domain adaptation methods within a mixture-of-experts ensemble system for nested NER from Arabic text, evaluated in a cross-domain setting. The system combines multiple Arabic pretrained encoders for token- and span-level classification with domain data augmentation, and integrates their predictions using a vote-based ensemble routing with a complementary XLM-R expert. For RE, we employ an adversarially trained relation classifier backed by auxiliary entity types, NLI reranking, and structural schema rules, treating `no_relation` as a standard relation label. We additionally study a positive-relation setting that excludes `no_relation` from both training and inference. Our NER system reaches 0.7438 micro F1 on the hidden test set. For RE, the positive-relation setting achieves 0.9608 micro F1, while the `no_relation`-aware system achieves 0.9295.

1 Introduction

Knowledge Graphs (KG) benefit myriad knowledge-aware systems such as search engines and dialogue systems that require accurate natural language understanding and complex reasoning (Ji et al., 2022). A KG contains entities as nodes and semantic relations between them as edges, forming a structured graph representation of knowledge (Hogan et al., 2021). Given increasing volumes of unstructured textual data, natural language processing methods for extracting entities and their relations support automated KG construction (Yan et al., 2018). However, there is limited research on methods for Arabic (Aljabari et al., 2026).

To bridge this gap, we address two tasks within the Arabic KG Construction and Domain Adaptation shared task (Aljabari et al., 2026): (1) nested named entity recognition (NER) spanning 21 semantic types from the WOJOOD schema (Jarrar et al., 2022) under a cross-domain setup, and (2) relation extraction (RE) following the WOJOODRELATIONS schema (Aljabari et al., 2025b).

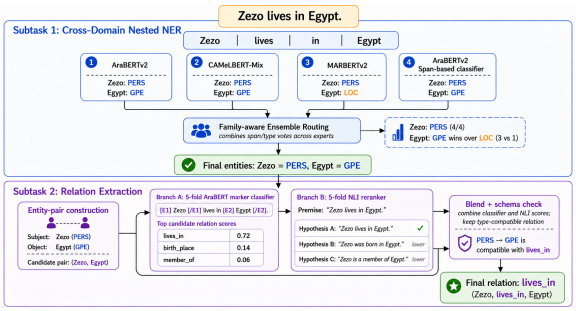


Figure 1: System inference.

Our system (Fig. 1) integrates data augmentation for developing an ensemble of heterogeneous NER experts across three pretrained Arabic encoders, whose outputs route through a trained XLM-R model providing additional candidate spans under a conservative rule. RE follows in a multistage pipeline consisting of a relation-existence classifier, followed for positive pairs by an adversarial- and auxiliary-trained relation classifier, a natural language inference (NLI)-based class ranker, and a relation-type validator. Our code is published at github.com/McMasterNLP/ArabicKG-2026.

2 Background

Domain adaptation addresses the challenge in which models suffer performance degradation on target domains that differ in vocabulary, style, or genre from the data seen in training (Han and Eisenstein, 2019; Gururangan et al., 2020), known to emerge in entity recognition (Lin and Lu, 2018; Jia et al., 2019) and relation extraction Bassignana and Plank (2022). Adaptation techniques can be applied at the data-level, such as controlled data augmentation and entity swaps that generate new

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training examples (Hu et al., 2023), and in training procedures, such as TAPT and DAPT that continue self-supervised masked language modeling (MLM) pretraining before downstream finetuning (Han and Eisenstein, 2019; Gururangan et al., 2020).

Toward *Arabic KG construction*, Jarrar et al. (2022) introduced the large multidialect Arabic nested NER corpus WOJOOD with 21 entity types and established multi-task flat-and-nested baseline architectures. Extensions included WOJOOD$_{fine}$ with finer-grained subtypes Liqreina et al. (2023), *WojoodOntology* which unifies external schemas (Aljabari et al., 2025a), and WOJOOD^{Relations} containing entity relation labels (Aljabari et al., 2025b). Hamad et al. (2025) introduced the multidialect NER dataset KONOOZ, containing several domains that are absent in WOJOOD. Their cross-domain evaluation of Arabic NER models revealed an F1-score performance degradation of up to 38% when transferring models trained on formal source data to unseen target domains. Motivated by these cross-domain challenges, we investigate several strategies within our Arabic knowledge graph construction pipeline for improving transfer to unseen domains, including encoder diversification, targeted data augmentation, span-based modeling, and ensemble routing.

3 System Overview

We propose a system that integrates several domain adaptation techniques for an Arabic KG construction pipeline (see Fig. 1). Given a raw Arabic input sentence, the system processes the data through two primary modules. First, a cross-domain NER ensemble detects and categorizes nested entities through three stages: (1) expert construction, (2) symmetric ensemble routing, (3) complementary candidate addition using XLM-R. Second, a multi-stage Relation Extraction (RE) pipeline first determines whether an entity pair expresses a relation or `no_relation`, then classifies positive pairs using the positive-relation pipeline.

3.1 Nested NER Expert Development

We develop four base experts for nested NER: three token-level models based on AraBERTv2, CAMeLBERT-Mix, and MARBERTv2 (Antoun et al., 2020; Inoue et al., 2021; Abdul-Mageed et al., 2021), and an AraBERTv2 span-based model following the span-classification approach of Elkhbir et al. (2023). We additionally train XLM-R base as a complementary token-level expert. The token-level models use BIO $\times$ entity-type labeling, while the span-level model predicts complete entity spans using boundary embeddings and span-width features. The Arabic encoder experts are trained on WOJOOD, with entity-swap augmentation applied to the primary token-level experts. The augmentation replaces selected isolated entities with WOJOOD entities of the same type while preserving the corresponding labels. For 10% of eligible training samples, isolated entity spans are swapped with spans of the same type from other samples–details and statistics in Table 2 and §B.

No KONOOZ sentences or labels are used for NER pretraining, supervised training, checkpoint selection, threshold calibration, or ensemble selection.

3.2 NER Ensembling Routing

Given an input sentence, the four primary experts independently produce NER predictions. We combine these predictions using a symmetric exact-span voting procedure.[1]

Candidate spans must receive at least two exact votes from the four base experts. Predicted spans of the same entity type that overlap in the text are grouped into components, and only candidates with the highest base-expert vote count are retained. When multiple candidates are tied, ties are resolved deterministically by preferring the shorter span, followed by earlier start position, and earlier end position. The selected spans form the ensemble baseline B.

XLM-R is used as a complementary expert after the primary ensemble has been fixed. We first construct the corresponding five-expert candidate set and identify XLM-R-supported spans that are introduced by the additional expert evidence. Only candidates that do not overlap an already accepted primary-ensemble span are added. Thus, XLM-R can add supported predictions but cannot replace or modify spans already selected by the four-expert ensemble.

3.3 Semantic Relation Extraction

Our RE system contains two stages. First, a relation-existence model determines whether an entity pair expresses a relation or `no_relation`. For pairs predicted as relations, an AraBERTv0.2-based relation classifier (RE-CLF) ranks the 40

[1] See §C for other ensemble approaches explored.

positive relation classes, the top five candidates are reranked by an NLI model, and a soft entity-schema penalty is applied to the resulting scores.

The entity types of relation arguments constrain the likelihood and validity of relation types–"*person* is *president of organization*" is valid, "*percent* is *president of website*" isn't. Thus, we train the RE-CLF in a multitask setup with auxiliary heads that learn to classify the subject and object entity types[2], all jointly optimized on a weighted objective: $L_{\text{total}} = L_{\text{relation}} + 0.15 \times (L_{\text{subject}} + L_{\text{object}})$, yielding entity-type-aware representations useful for relation classification. We perform adversarial training, using fast gradient method (FGM), wherein after a forward pass on an input, another forward pass is performed on a new adversarial input created by perturbing the input embeddings toward the gradient (Goodfellow et al., 2015; Miyato et al., 2017).

To model `no_relation`, we train a separate AraBERTv0.2 model on all 17,367 cleaned source instances with a binary relation-existence head and a 40-way positive-relation head, together with the auxiliary subject and object entity-type heads. The existence loss is computed over all training instances, while the 40-way relation loss is computed only for positive instances. FGM adversarial training and the auxiliary subject and object entity-type objectives are retained. At inference, the relation-existence probabilities from the five folds are averaged and a fixed threshold of 0.5 is applied. Pairs below the threshold are assigned `no_relation`; the remaining pairs are passed to the positive-relation pipeline described above.

Inspired by Aljabari et al. (2025b), we add a module that reranks RE-CLF relation candidates with NLI model to resolve ambiguity between semantically similar relations (e.g., `birth_place` & `lives_in`). Treating the original text as a premise, hypotheses are constructed by slotting a (subject, object, relation) into natural language templates[3], as in "{subject} was born in {object}". For the NLI model, AraBERTv0.2 is finetuned on constructed (premise, hypothesis) pairs from the training data, with *entailment* labels on hypotheses with ground truth relations, and *contradiction* labels on samples created by swapping semantically similar (not same) entities or relations. At inference time, we run the NLI model for the top-5 relation types, and rerank them by entailment likelihood.

[2]Entity labels obtained from our NER system (see §D.3).

[3]We use template structures from SinaLab (2025)

Method	F1	Pr	Re
Four-expert ensemble + XLM-R	**0.7438**	**0.75**	**0.7378**

Method	OOF F1	Blind test
Positive-relation pipeline	0.9715	0.9608
`no_relation`-aware system	0.9242	0.9295

Table 1: NER results on the official KONOOZ test set (top) and RE results (bottom). For RE, the positive-relation OOF result is evaluated over positive relation instances, while the `no_relation`-aware OOF result uses the complete sentence-grouped OOF data, including negative pairs.

The third module derives final scores S for a candidate r. S_1 aggregates RE-CLF and NLI probabilities as $S_1(r) = 0.55 \cdot P_{RE}(r) + 0.45 \cdot P_{NLI}(r)$. The r is checked against a schema[4] of compatible relation- and argument-entity types $\mathcal{C}$. Invalid types can result from NER errors, so rather than eliminating such candidates, S_2 integrates a penalty:

$$S_2(r) = \begin{cases} S_1(r), & r \in \mathcal{C}, \\ S_1(r)\exp(-\lambda), & r \notin \mathcal{C}, \end{cases} \quad \lambda = 2.0$$

4 Experimental Setup

The final NER evaluation measures precision, recall, and micro F1 across ten unseen target domains on a hidden test set from the KONOOZ corpus, to assess cross-domain generalization. For an out-of-domain development benchmark, we experiment on CV-18 (Saidi et al., 2026), an external public Arabic NER dataset annotated with the same scheme. RE systems are developed and evaluated on the formal train/test splits of WOJOOD^{Relations}, which contains 40 positive relation labels together with `no_relation`, with micro F1 as the primary performance metric. We report both the positive-relation pipeline and a `no_relation`-aware configuration. The latter uses a source-trained relation-existence model with a fixed decision threshold of 0.5, without using hidden-test predictions to select or adjust the threshold.[5]

All token-level NER models and AraBERT-based RE models are optimized using AdamW. All hyperparameters and training configs are in §B.

5 Results and Discussion

Table 1 summarizes the NER and RE results from the competition. Our final NER system, developed without KONOOZ pretraining or adaptation, achieved a micro F1 of 0.7438 on the hidden KONOOZ test set, with 0.7500 precision and 0.7378 recall.

For RE, the positive-relation pipeline combining auxiliary NER learning, FGM adversarial training, NLI reranking, and schema constraints achieved 0.9608 micro F1 on the blind test, with no exact-match overrides applied. Adding explicit `no_relation` handling through the source-trained relation-existence model with a fixed 0.5 threshold resulted in 0.9295 micro F1. We discuss both settings below.

5.1 NER Analysis

Further metrics on both the auxiliary CV-18 benchmark and KONOOZ test set are in §C, including additional systems explored in development.

Auxiliary Out-of-Domain Evaluation. We computed Spearman rank correlation between micro F1 performances of 14 systems evaluated on both CV-18 and the KONOOZ hidden test set. System ranking across the two datasets show a moderate positive correlation (Spearman $\rho = 0.6535$, $p = 0.011$, $N = 14$). This shows that while CV-18 performance is not a perfect proxy for KONOOZ performance, it provides a practical out-of-domain benchmark for system development. Therefore, we base our system design decisions on CV-18 performance rather than performance gains on the in-domain development set.

Expert Complementarity. We analyze exact-span agreement among AraBERT (A), CAMeLBERT (C), MARBERTv2 (M), and the AraBERT span-based model (S) on CV-18, to determine whether ensemble reliability depends only on the number of supporting experts or also on the architectural diversity of those experts. On CV-18 development data, candidate reliability increases strongly with agreement among the four primary experts. Predictions supported by all four experts achieve 95.1% precision, compared with 84.0% for three-expert agreement, 59.7% for two-expert agreement, and 32.4% for single-expert predictions. This supports requiring agreement from at least two experts in the primary ensemble. This supports the use of architecturally complementary experts, with the span-based model providing particularly useful boundary information.

Routing and Complementary XLM-R Predictions. Using the clean source-only experts, the symmetric four-expert ensemble achieved 0.7780 micro F1 on the frozen CV-18 test set. We then evaluated XLM-R as a complementary expert. The addition policy was selected using CV-18 development data and does not allow XLM-R to modify spans already accepted by the four-expert ensemble. Adding the resulting non-overlapping XLM-R-supported candidates increased CV-18 test micro F1 to 0.7843. Applying the same frozen procedure to the hidden KONOOZ test set produced 0.7438 micro F1, with 0.7500 precision and 0.7378 recall.

Boundary Errors. NER errors frequently involve boundary ambiguity. NeRValuate analysis (Batista and Upson, 2026) of an earlier ensemble configuration on CV-18 identifies 124 boundary-only errors and 26 errors involving both boundary and entity type, meaning that 150 of 215 aligned mismatches (69.8%) involve incorrect entity boundaries. These include cases where predictions include only part of a multi-token entity, fragment one entity into multiple spans, or incorrectly merge adjacent entities. For example, in an expression corresponding to "an hour ago", AraBERT predicts only "hour" as TIME, while another expert recovers the complete gold span. Conversely, an expert can incorrectly merge two adjacent PERS entities into a single span. These cases motivate our use of complementary expert evidence for boundary decisions.

5.2 RE Analysis

Semantic Resolution. We initially submitted an RE-CLF AraBERT baseline which reached 0.9535 micro F1 on the blind test set. On development data, we observed its errors often involved confusion between semantically similar relations such as `birth_place` & `lives_in`; quantitatively, the highest-likelihood class matched the ground truth in 96.59% of cases, while it was among the top five in 99.61%, which were typically semantically similar. This motivated NLI reranking.

Dependency Between Pipeline Stages. Introducing FGM and auxiliary NER learning changed the primary classifier's error distribution, reducing the NLI reranker performance trained on the original classifier's errors. Retraining the reranker on hard

[4] Schema derived from training data–see §D.6.

[5] Detailed dataset statistics and split breakdowns are provided in Table 3; see §E for evaluation of negative relations.

negatives from the updated classifier improved the combined system, suggesting downstream components may need to be retrained when upstream changes substantially alter their error distributions.

Schema constraints provide targeted corrections but remain imperfect: on OOF predictions, they changed 51 of 9,289 decisions, correcting 31 errors while introducing 16 new ones, supporting their use as soft rather than hard constraints.

Handling `no_relation`. To explicitly model negative entity pairs, we use the relation-existence head of the two-head AraBERT model to gate the positive-relation pipeline. Using the fixed 0.5 threshold, the relation-existence decision achieves 0.9510 recall for relation instances and 0.9308 recall for `no_relation` in sentence-grouped five-fold OOF evaluation, corresponding to 0.9409 balanced accuracy. When combined with the positive-relation pipeline, the system achieves 0.9242 positive-class micro F1 over the complete OOF set and 0.9300 full 41-class accuracy. The corresponding blind-test result is 0.9295 micro F1. The threshold is fixed from the source-side setup and is not adjusted using hidden-test predictions.

6 Conclusion

We addressed cross-domain Arabic knowledge graph construction across the unseen domains of the KONOOZ corpus. For nested NER, combining token-level models from different encoder families with an AraBERT span-based expert provided complementary evidence under domain shift. Symmetric routing over four source-trained experts followed by complementary XLM-R additions achieved 0.7438 micro F1 on the hidden KONOOZ test set without KONOOZ pretraining or adaptation. For RE, the positive-relation pipeline without exact-match overrides achieved 0.9608 micro F1 on the blind test. When extended with explicit `no_relation` prediction with a fixed 0.5 threshold, the system achieved 0.9295 micro F1. Overall, the results demonstrate the value of combining complementary model architectures with conservative ensemble decisions for cross-domain Arabic KG construction.

Limitations

NER development also partly relies on CV-18, which does not fully represent the linguistic and domain variation of the hidden KONOOZ data, as reflected by the moderate correlation between their system rankings. Finally, annotation inconsistencies in the WOJOOD nested data introduce training and evaluation noise and limited the reliable use of automatically generated augmentation examples.

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Appendix

A Dataset Statistics

Table 2: Dataset statistics for the WOJOOD source corpus and 10% entity-swap augmentation setup.

Split / Setup	Sentences	Tokens	Entity Types
Train (Raw Wojood)	23,125	390,999	–
10% Entity-Swap Augmented	1,341	24,057	–
Validation	3,304	57,555	–
Test	6,606	113,725	–
Total Source Corpus (Raw)	**34,376**	**586,336**	**21**

Table 3: Dataset statistics for the WOJOOD^{Relations} dataset across splits, detailing data pre-processing modifications.

Split / Category	Raw Released	Processed Setup
Training Triples	**17,381**	**17,367**
Positive Relations	9,289	9,289
Negative (`no_relation`)	8,092	8,078
Contradictory Rows Removed	0	14
Validation Triples	**2,074**	**2,074**
Inspected / Analyzed Rows	2,074	2,074
Blind Test Triples	**4,386**	**4,386**
Gold label distribution	Not released	

B Training and Reproducibility Details

B.1 Token-Level NER Training

The token-level NER experts use the ArabicNER nested tagging pipeline with 21 BIO classification heads. AraBERTv2, CAMeLBERT-Mix, and MARBERTv2 are fine-tuned on the clean WOJOOD nested training data. AdamW is used for optimization, and corrected runs use validation micro F1 for checkpoint selection.

B.2 Entity-Swap Augmentation

We applied entity-swap augmentation to 10% of the 13,408 eligible WOJOOD training sentences. With seed 42, 1,341 sentences were selected and one isolated entity span was replaced in each sentence by another entity of the same type drawn from the WOJOOD training data, while preserving the corresponding labels.

The augmentation is restricted to DATE, GPE, ORG, and PERS entities. Replacement candidates match the entity type and token length; for multi-token GPE and ORG entities, replacements additionally preserve the entity head token. The augmented examples are appended to the original 23,125-sentence training set, producing 24,466 sentences and 415,056 tokens.

The final entity-swap experts are obtained by continuing from their selected clean-WOJOOD checkpoints for one epoch using learning rate 5×10^{-6}, batch size 8, and seed 42. Development and test splits remain unchanged.

B.3 AraBERT Span Classifier

To provide architectural diversity beyond BIO token classification, we train an AraBERTv2 multi-label span classifier following Elkhbir et al. (2023). First-subtoken encoder representations are contextualized using a single-layer bidirectional LSTM. Each candidate span is represented using its start and end states together with a learned 64-dimensional span-width embedding. The resulting span representation is passed through a two-layer feed-forward projection before independent classification over the 21 entity types.

The model enumerates spans up to 15 tokens and is trained using weighted binary cross-entropy, with negative span–label pairs downweighted by a factor of 0.02. The encoder uses learning rate 1×10^{-5}, while the BiLSTM, span-width representation, projection layers, and classifier use learning rate 1×10^{-4}. Training uses batch size 4, seed 42, 5% warmup, and gradient clipping at 1.0. The final expert uses the continued epoch-8 checkpoint. Entity-specific decision thresholds are calibrated during development before final inference.

B.4 XLM-R Complementary Expert

The complementary XLM-R model is fine-tuned on the clean WOJOOD nested training data using XLM-R-base. The model uses the same 21-head BIO formulation as the token-level Arabic encoders.

Checkpoints are saved after every supervised epoch, and checkpoint 2 is selected using CV-18 development performance.XLM-R does not participate in the primary four-expert vote. It is introduced after the primary ensemble has been fixed and is used only to provide additional non-overlapping candidate spans under the complementary addition procedure described in Section 3.2.

Setting	Value
Encoder	XLM-R-base
Learning rate	2×10^{-5}
Batch size	8
Maximum epochs	20
Patience	4
Maximum length	512
Dropout	0.1
Gradient clip	1.0
Seed	42
Optimizer	AdamW

Table 4: Training configuration for the complementary XLM-R expert.

Method	Micro F1	Macro F1	Weighted F1	Pr	Re	Acc.
Four-expert ensemble + complementary XLM-R	**0.7438**	**0.6530**	**0.7381**	**0.7500**	**0.7378**	**0.9950**

Table 5: NER performance on the hidden KONOOZ test set. The system is developed without KONOOZ pretraining or adaptation.

Method	Micro F1
Symmetric four-expert ensemble	0.7780
+ complementary XLM-R	**0.7843**

Table 6: NER ensemble results on the frozen CV-18 test set. Ensemble decisions are selected using CV-18 development data.

C NER System Experiments

C.1 Ensembles

Our final NER system combines four primary experts based on AraBERTv2, CAMeLBERT-Mix, MARBERTv2, and an AraBERTv2 span-based classifier. Their predictions are first combined using symmetric exact-span voting. XLM-R is then used as a complementary expert to add supported predictions that do not overlap spans already accepted by the four-expert ensemble.

The Symmetric Four-Expert Ensemble combines predictions from AraBERTv2, CAMeLBERT-Mix, MARBERTv2, and the AraBERTv2 span-based model. A candidate span requires at least two exact expert votes. For overlapping predictions of the same entity type, candidates with the highest expert support are retained; remaining ties are resolved by preferring the shorter span, followed by the earlier span.

Complementary XLM-R Expert XLM-R is trained independently on WOJOOD and does not participate in the primary four-expert vote. After the primary ensemble predictions are fixed, XLM-R is used to identify additional supported candidate spans. Only additions that do not overlap an already accepted primary-ensemble span are retained. Thus, XLM-R can add predictions but cannot replace predictions already selected by the four-expert ensemble. All NER experts in this configuration are developed without KONOOZ pretraining or adaptation. Model and routing decisions are made using WOJOOD and the auxiliary CV-18 development benchmark. Table 5 reports the performance of the final NER system on the official KONOOZ hidden test set. Table 6 reports the contribution of the ensemble components on CV-18, which is used as the auxiliary out-of-domain benchmark during development.

C.2 Expert Support Patterns

Table 7 reports the reliability of exact-span predictions according to the number of primary experts supporting them. Agreement among multiple experts is substantially more reliable than single-expert predictions, supporting the use of a minimum two-expert agreement in the primary ensemble.

Supporting experts	Precision
4 of 4	0.9509
3 of 4	0.8398
2 of 4	0.5971
1 of 4	0.3241

Table 7: Exact-span precision on CV-18 development data according to the number of supporting primary experts.

C.3 Auxiliary Target-Adaptation Experiment

Our main NER result assumes that unlabeled target-domain text is not available before model development, consistent with the cross-domain setting of the shared task. In practice, however, some applications may provide access to unlabeled target-domain text before inference. We therefore also investigated target-adaptive pretraining as an auxiliary setting. A target-adapted configuration reached 0.7499 micro F1 on KONOOZ, compared with 0.7438 for the final system reported above. We retain the non-adapted system as our main result because it does not assume prior access to the target corpus, while the auxiliary result shows the potential benefit of unlabeled target-domain data when such access is available.

D Relation Extraction Training and Development Analysis

D.1 Training Hyperparameters

The relation classifier and NLI reranker both use AraBERTv0.2. Table 8 summarizes their main training configurations.

Setting	RE classifier	NLI reranker
Encoder	AraBERTv0.2	AraBERTv0.2
Learning rate	2×10^{-5}	2×10^{-5}
Batch size	16	8 groups
Maximum epochs	5	4
Patience	2	2
Maximum length	384	384
Weight decay	0.01	0.01
Warmup ratio	0.10	0.10
Seed	42	42

Table 8: Main training hyperparameters for Subtask 2.

Five sentence-grouped folds are trained independently. Sentence grouping ensures that entity pairs originating from the same sentence are not split across training and validation folds. At inference, the probabilities produced by the five fold-specific classifiers are averaged.

The `no_relation`-aware model uses the same sentence-grouped five-fold structure. At inference, the five fold-specific relation-existence probabilities are averaged before applying the fixed 0.5 decision threshold.

D.2 Marker-Based Relation Classifier

The positive-relation classifier is a 40-way AraBERTv0.2 classifier over the positive relation labels. We first preprocess the sentences by inserting `[SUBJ]` and `[OBJ]` token markers around the entities in the target subject-object pairs (Baldini Soares et al., 2019), as in "`[SUBJ] Zezo [/SUBJ] lives in [OBJ] Egypt [/OBJ]`".

This marker-based AraBERT classifier provides the base positive-relation probabilities used by the NLI reranker and schema adjustment. Explicit `no_relation` prediction is handled by the relation-existence component described in Section E.

D.3 Entity Type Training Labels

We obtain entity type labels for the RE dataset using our NER system; it identified entity labels for both relation arguments in 86.07% of training instances. We account for the missing labels during training by masking out the auxiliary losses so that only the relation loss contributes.

D.4 Fast Gradient Method Adversarial Training

We additionally apply Fast Gradient Method (FGM) adversarial training to improve the robustness of the AraBERT relation classifier. FGM is applied to the model's input embedding parameters during training; it does not modify the Arabic sentence or generate new textual training examples.

Given the gradient g of the clean training loss with respect to an embedding parameter θ, the adversarial perturbation is

$$r_{\text{adv}} = \epsilon \frac{g}{\|g\|_2},$$

and the perturbed parameter becomes

$$\theta_{\text{adv}} = \theta + r_{\text{adv}}.$$

We use $\epsilon = 1.0$. For each training batch, gradients are first computed on the original input. The embedding parameters are then perturbed in the gradient direction and an additional adversarial forward and backward pass is performed. The perturbation is removed before the optimizer update. The adversarial loss is weighted by $\lambda_{\text{adv}} = 1.0$:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{multi}} + \lambda_{\text{adv}} \mathcal{L}_{\text{adv}}.$$

This training procedure encourages the encoder to remain stable under small embedding-space perturbations. Adding FGM increases mean OOF F1 from 0.9612 to 0.9643, and combining it with the auxiliary entity-type objective further increases F1 to 0.9659.

D.5 NLI Reranker

For each candidate relation, a fixed relation-specific template is used to construct a natural-language hypothesis containing the subject and object. The original sentence acts as the premise, and the NLI model estimates whether the sentence supports the candidate relation. This formulation allows the reranker to compare the meanings of semantically similar relation labels directly against the source sentence.

During training, each NLI group contains one positive relation hypothesis and five hard-negative hypotheses selected from competing relation predictions. The NLI objective combines binary entailment classification with a ranking loss weighted by 0.50.

We retain the five-fold development structure for the NLI stage, producing five fold-specific

rerankers. At inference, each of the relation classifier's top five candidate relations is scored by all five rerankers and their scores are averaged. The averaged NLI scores are converted to probabilities using temperature 0.75.

$$p_{\text{blend}}(r) = 0.55\, p_{\text{RE}}(r) + 0.45\, p_{\text{NLI}}(r).$$

The schema adjustment described below is applied only after this semantic score blending.

D.6 Entity-Schema Audit and Soft Penalty

The entity types produced by Subtask 1 provide an additional source of evidence about which relations are plausible for a subject–object pair. Because NER predictions can themselves be incorrect, we do not use entity types as hard constraints. Instead, we first audit relation–type compatibility on the training data and then use the resulting schema as a soft adjustment.

For each positive relation with known subject and object types, we compare the observed ordered entity-type pair with the proposed subject–object type configuration for that relation. Training instances are categorized as matching the expected direction, matching the reverse direction, or matching neither direction. A relation is retained as schema-active only when it has at least five auditable training examples and at least 80% type compatibility.

This audit produces 30 active relation types. It also identifies relation-specific inconsistencies in the training data. In particular, the operational direction of `GPE.has_population` is reversed to reflect the observed `CARDINALITY`→`GPE` pattern. The relations `Administration.leader_of`, `GPE.capital_of`, and `GPE.has_currency` are excluded from the active schema because their observed subject–object type patterns are not sufficiently consistent.

At inference, the schema adjustment defined in §3.3 is used only when reliable entity types are available for both the subject and object.

Incompatible candidates are not removed from consideration. This is important because an incorrect NER prediction should not be able to eliminate a relation that is strongly supported by the sentence. The final schema is applicable to 3,806 of the 4,386 blind-test instances, where sufficiently reliable subject and object types are available.

D.7 Component Ablation

Table 9 shows the progressive development of the positive-relation component using sentence-grouped five-fold out-of-fold evaluation over the positive relation instances. We report the blind-test submission progression separately in Table 10, since not every development configuration was submitted to the shared-task server.

Method	Mean OOF F1
AraBERT + markers	0.9612
+ FGM	0.9643
+ auxiliary NER heads	0.9659
+ retrained NLI blend	0.9701
+ schema penalty	**0.9715**

Table 9: Progressive development of the positive-relation component using sentence-grouped five-fold out-of-fold evaluation over positive relation instances.

RE configuration	Blind-test F1
AraBERT marker ensemble	0.9535
+ NLI reranking	0.9580
+ schema penalty	0.9583
FGM + NER-aux + retrained NLI + schema	**0.9608**
+ `no_relation` prediction ($\tau = 0.5$)	0.9295

Table 10: Subtask 2 blind-test results. The first four configurations classify among the positive relation labels. The final configuration additionally predicts `no_relation` using the source-trained relation-existence model with a fixed 0.5 threshold.

D.8 Reranker Adaptation

The initial NLI reranker was developed using errors produced by the earlier AraBERT relation classifier. After FGM adversarial training and auxiliary entity-type supervision were introduced, the primary classifier produced a different distribution of mistakes. In particular, the identities and rankings of the hard negative relations changed.

We therefore rebuilt the NLI training data using hard negatives produced by the updated FGM + NER-auxiliary classifier and retrained all five NLI rerankers. The resulting NLI training set contains 65,023 binary premise–hypothesis instances constructed from 9,289 positive RE rows, including 18,578 entailment and 46,445 non-entailment examples.

With the retrained reranker, the final blend uses weights of 0.55 for the AraBERT relation classifier and 0.45 for NLI, with temperature 0.75. This configuration reaches 0.9701 mean OOF F1 before

the schema-level adjustment. Applying the audited schema increases the positive-relation OOF score to 0.9715 and changes 29 blind-test predictions.

E Negative Relations in Subtask 2

The task permits either one of the 40 positive relation labels or `no_relation` for an entity pair. After removing 14 contradictory negative instances, the cleaned source training data contains 17,367 instances, including 9,289 positive relations and 8,078 `no_relation` instances.

To model this distinction explicitly, we train an AraBERTv0.2 model with a shared encoder and two classification heads. A binary relation-existence head predicts whether the entity pair expresses a relation, while a 40-way relation head predicts the positive relation class. The relation-existence loss is computed over all training instances, whereas the 40-way relation loss is computed only for positive instances. FGM adversarial training and the auxiliary subject and object entity-type objectives are retained.

At inference, the relation-existence probabilities from the five fold-specific models are averaged. A fixed threshold of 0.5 is applied to the resulting relation probability. Instances below this threshold are assigned `no_relation`. For the remaining instances, the positive relation is produced by the FGM + auxiliary-NER classifier, NLI reranking, and soft schema adjustment described in Section 3.3. No hidden-test predictions are used to select or adjust the relation-existence threshold.

System	Rel. Re	no_relation Re	Bal. Acc.	Positive F1	41-class Acc.
no_relation-aware RE ($\tau = 0.5$)	0.9510	0.9308	0.9409	0.9242	0.9300

Table 11: Sentence-grouped five-fold OOF evaluation of the `no_relation`-aware RE system over the complete 17,367-instance source data. Relation and `no_relation` recall and balanced accuracy evaluate the relation-existence decision; positive F1 and 41-class accuracy evaluate the resulting RE predictions.

The source OOF results show that the relation-existence model retains high recall for both positive relations and `no_relation`. When combined with the positive-relation pipeline, the corresponding blind-test configuration achieves 0.9295 micro F1. The positive-relation experiment, which excludes `no_relation` from training and inference, achieves 0.9608 micro F1 on the same blind test.